

SOHAM ASHISH GAONKAR

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RESEARCH INTERESTS

Computer Vision, 3D Scene Reconstruction (Gaussian Splatting, Neural Rendering), Efficient Machine Learning (Model Compression, Quantization, Efficient Inference), Systems for ML/AI, Medical Image Computing.

EDUCATION

Indian Institute of Technology Gandhinagar , Gujarat, India	July 2023 – Present
<i>Bachelor of Technology in Artificial Intelligence</i>	CPI: 9.3 / 10.00 (Rank: 1st)
<i>Graduate Status: Admitted to M.Tech. Research at Indian Institute of Science (IISc), Bangalore (Early Selection)</i>	
Pace Junior Science College , Mumbai, India	2022 – 2023
<i>Class XII (HSC Board, Maharashtra)</i>	Score: 87.0%
Pawar Public School , Mumbai, India	2012 – 2021
<i>Class X (ICSE Board)</i>	Score: 98.2%

PUBLICATIONS AND MANUSCRIPTS

- [1] **S. Gaonkar**, H. Shekhar, K. Bader, et al. “Deep Learning-based Ultrasound Image Segmentation for Histotripsy Ablation Zone Delineation.” *IEEE International Ultrasonics Symposium (IUS 2025)*, Rotterdam, Netherlands. (Accepted)
- [2] **S. Gaonkar**, H. Shekhar, K. Bader, et al. “Improved Segmentation of Histotripsy Bubble Clouds Using Modern CNN Architectures and Combined Dice–Focal Loss.” *Frontiers in Acoustics*. (Submitted)

Note on Co-authors: All authors are listed in their order of contribution. If additional collaborators or co-authors contributed to these manuscripts, they should be added in full sequence here to ensure academic transparency.

RESEARCH EXPERIENCE

Samsung Research Institute, Noida May 2026 – Present
Research Intern — Next Level Labs, On-device AI Team

- Optimizing Large Vision-Language Models (VLMs) and LLMs for on-device inference on Samsung mobile chipsets, focusing on memory-efficient attention layouts and quantization-aware compression.
- Designing end-to-end applications integrating on-device multimodal models for real-time image understanding.
- Performing detailed latency profiling of transformer attention layers on mobile NPUs/GPUs, resolving tensor layout and token generation latency issues.

Soket AI Jan 2026 – Feb 2026
Research Intern

- Worked on Agri-LLM, developing a language model designed to serve Indian agricultural applications.
- Developed LLM-as-a-judge evaluation frameworks and synthesized supervised fine-tuning (SFT) data to improve response reliability.

CVIG Lab, IIT Gandhinagar [Code] | May 2025 – July 2025
Summer Research Intern (Advisor: Prof. Shanmuganathan Raman; Collaborators: Dr. Ashish Tiwari, Prajwal Singh)

- Designed *PanoTo3D*, a pipeline for single-panorama 3D scene reconstruction and novel view synthesis.
- Combined monocular depth estimation with RGB panoramas to form point clouds; synthesized multi-view supervision via inpainting to stabilize geometry learning; trained 3D Gaussian Splatting models for sparse-view inverse rendering. [Interactive Demo]

MUSE Lab, IIT Gandhinagar [Code] | Jan 2025 – April 2025
Undergraduate Researcher (Advisor: Prof. Himanshu Shekhar; Collaborator: Prof. K. Bader, UChicago)

- Engineered DeepLabV3-based segmentation models with custom combined Dice–Focal loss to identify histotripsy ablation bubble clouds in ultrasound imagery. Raised mean IoU to **83%** (+7% over baseline) and accuracy to **97%** on low-contrast clinical datasets.

SELECTED RESEARCH PROJECTS

LLM Compression via Neural Alignment & Layer Merging [Code] | Oct 2025

Advisor: Prof. Mayank Singh, IIT Gandhinagar

- Investigated representation alignment limits in LLaMA-3 (8B), analyzing task-dependent representation decay.
- Optimized layer merge weights via gradient descent and Bayesian search; implemented Hungarian permutation matching for optimal neuron alignment, achieving +0.0864 MMLU score improvement at 12.5% parameter compression.

SWaT Anomaly Detection & Root Cause Analysis [Code] | Jan 2026

Industrial Project IIT Gandhinagar

- Engineered a 15-stage LSTM Autoencoder pipeline to detect structural anomalies on the Secure Water Treatment (SWaT) industrial testbed dataset. Integrated Bayesian Networks for causal modeling and reconstruction-based root cause analysis (RCA).

Lightweight Autograd Framework (miniTorch) [Code] | June 2025

Independent Project

- Programmed a NumPy-only deep learning library from scratch supporting reverse-mode autograd, dynamic computation graphs, broadcasting, neural layers, and SGD/Adam optimizers.

JPEG Compression Pipeline on FPGA [Code] | April 2025

Advisor: Prof. Joycee Mekie, IIT Gandhinagar

- Designed synthesizable Verilog modules and finite state machines (FSMs) on Basys3 FPGA for discrete cosine transform (DCT) and coefficient pruning, achieving 50% compression with PSNR > 30 dB.

ACADEMIC HONORS AND AWARDS

- Academic Scholarships & Ranks:** Academic Excellence Scholarship (AY 2024–25, AY 2025–26); Department Rank 1 (B.Tech. AI, IITGN).
- National Rank:** JEE Advanced AIR 2825 (Top 1.5% of 190k candidates); JEE Main AIR 1987 (Top 0.2% of 1M candidates).
- Undergraduate Research Showcase Award (2nd Rank):** For outstanding medical image segmentation research (IITGN).
- Competitions:** Top 150 teams (out of 20,000+) in Amazon ML Challenge 2025; Robotics Runner-Up at HackRush (IITGN Hackathon).

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Verilog, JavaScript, Java, SQL
Libraries & Frameworks: PyTorch, Hugging Face Transformers, OpenCV, Scikit-learn, Weights & Biases, SciPy
Systems & Tools: Git, Docker, Basys3 FPGA, Xilinx Vivado, VS Code, Linux, FastAPI, React

RELEVANT COURSEWORK

Computer Science & AI: Machine Learning [A], Computer Vision [A], Natural Language Processing [A-], Introduction to Data Science [A], Foundations of AI: Multiagent Systems [B], Databases [A-], Operating Systems [B], Software Tools & Techniques for AI [A], Computing [A+], Data Structures & Algorithms [A-], Theory of Computing [A-].
Mathematics: Mathematical Foundations for AI [A], Calculus & Linear Algebra [A], Ordinary Differential Equations [A+], Signals, Systems, & Random Processes [A], Introduction to Partial Differential Equations [A], Calculus of Several Variables [A-].

TEACHING & ACADEMIC SERVICE

Calculus & Linear Algebra (MA 103) Academic Mentor Sept 2024 – Nov 2024

Academic Development & Honors (ADH) Cell, IIT Gandhinagar

- Conducted weekly problem-solving tutorials and conceptual clarifications for a class of over 300 undergraduate students.
- Collaborated with the math department faculty to review homework questions and design targeted worksheets.

Machine Learning Club Core Member Sept 2024 – April 2025

IIT Gandhinagar

- Mentored junior undergraduate students on core machine learning libraries (NumPy, Pandas, PyTorch) and research workflows.